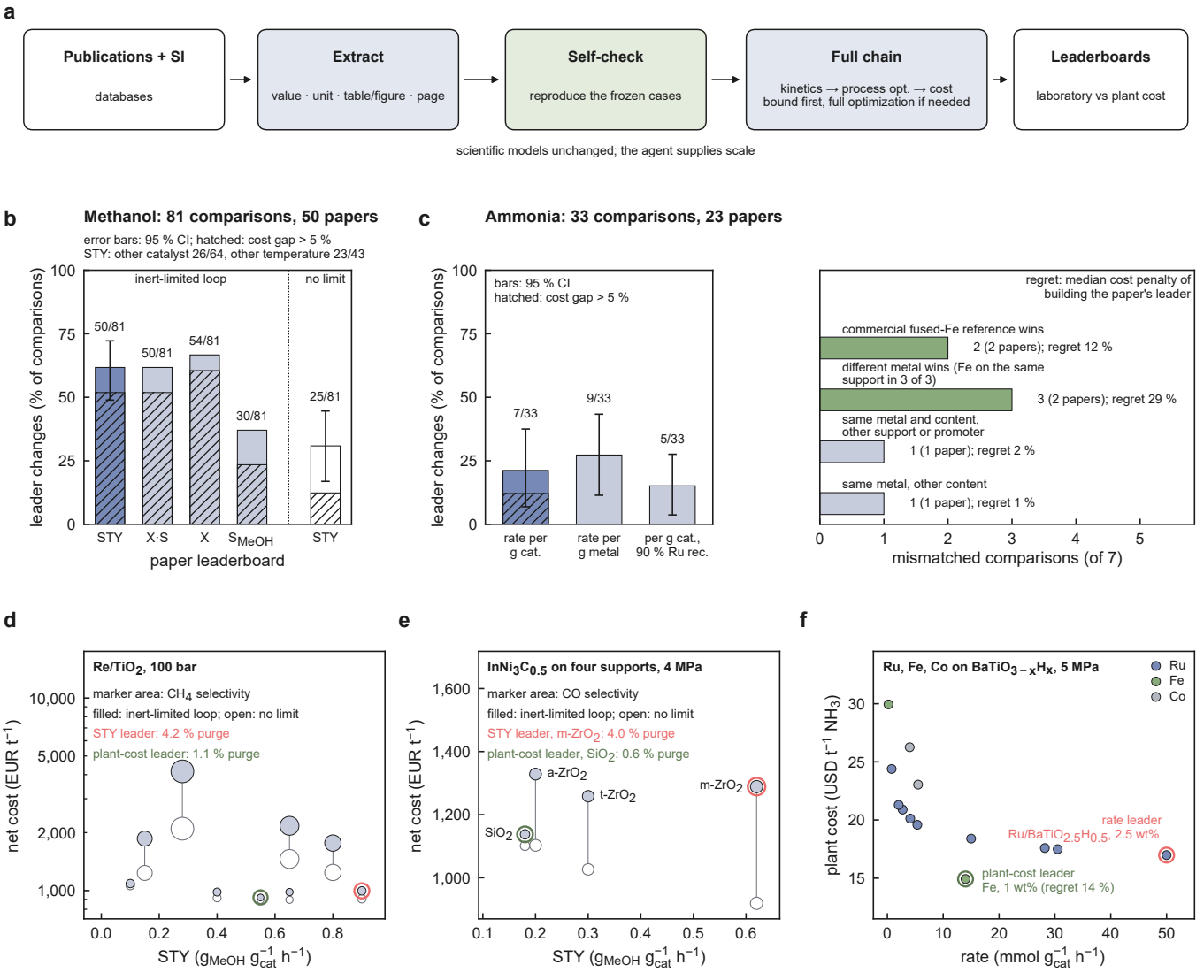

a
Publications + SI
databases
Extract
value · unit · table/figure · page
Self-check
reproduce the frozen cases
Full chain
kinetics → process opt. → cost
bound first, full optimization if needed
Leaderboards
laboratory vs plant cost
scientific models unchanged; the agent supplies scale
b
Methanol: 81 comparisons, 50 papers
error bars: 95 % CI; hatched: cost gap > 5 %
STY: other catalyst 26/64, other temperature 23/43
inert-limited loop
no limit
leader changes (% of comparisons)
50/81
50/81
54/81
30/81
25/81
STY
X·S
X
S_MeOH
STY
paper leaderboard
c
Ammonia: 33 comparisons, 23 papers
bars: 95 % CI
hatched: cost gap > 5 %
7/33
9/33
5/33
rate per g cat.
rate per g metal
per g cat., 90 % Ru rec.
regret: median cost penalty of building the paper's leader
commercial fused-Fe reference wins
2 (2 papers); regret 12 %
different metal wins (Fe on the same support in 3 of 3)
3 (2 papers); regret 29 %
same metal and content, other support or promoter
1 (1 paper); regret 2 %
same metal, other content
1 (1 paper); regret 1 %
mismatched comparisons (of 7)
d
Re/TiO2, 100 bar
marker area: CH4 selectivity
filled: inert-limited loop; open: no limit
STY leader: 4.2 % purge
plant-cost leader: 1.1 % purge
net cost (EUR t−1)
STY (gMeOH gcat−1 h−1)
e
InNi3C0.5 on four supports, 4 MPa
marker area: CO selectivity
filled: inert-limited loop; open: no limit
STY leader, m-ZrO2: 4.0 % purge
plant-cost leader, SiO2: 0.6 % purge
a-ZrO2
t-ZrO2
m-ZrO2
SiO2
net cost (EUR t−1)
STY (gMeOH gcat−1 h−1)
f
Ru, Fe, Co on BaTiO3−xHx, 5 MPa
Ru
Fe
Co
rate leader
Ru/BaTiO2.5H0.5, 2.5 wt%
plant-cost leader
Fe, 1 wt% (regret 14 %)
plant cost (USD t−1 NH3)
rate (mmol gcat−1 h−1)